

Re-verifying the Herglotz function at real quadratic units: two errata in Radchenko–Zagier’s tables, and the elementary boundary

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2026-06-25

Research report. Created (UTC) 2026-06-25T04:05:20Z; repository HEAD
210c21f70bdd0023b765a66a61a3aa04fd34c550. Store `herglotz-quadratic-units.wl`.
Source: D. Radchenko and D. Zagier, *Arithmetic properties of the Herglotz function*
(`src/PolyLog/docs/pdfs/HerglotzFunction/`). Reproducer:
`src/PolyLog/tools/scratch/herglotz/verify_full.gp`.

Abstract

We re-implemented the Herglotz function and its companion $\mathcal{J}(x)$ independently (PARI/GP and `mpmath`), verified the paper’s defining equations to $\sim 10^{-20}$, and then re-checked *every* row of Radchenko–Zagier’s Tables 1 and 2 of one-class-per-genus values $\mathcal{J}(n + \sqrt{n^2 \pm 1})$ at 200-digit precision. Forty-three of the forty-five rows match to better than 10^{-80} . The two that do not are typographical errors in the printed paper: in Table 1, the $n = 43$ row carries a spurious extra term $+S_{33,56}$, and the $n = 61$ row prints $S_{60,1032}$ where it should print $S_{60,248}$. Each correction is the *unique* single edit (subscript change, coefficient change, or term deletion) that restores the identity; with both applied, all forty-five rows verify to 10^{-80} . As a by-product we map the boundary of elementary evaluability: $\mathcal{J}(x)$ closes in $\{\zeta(2), \log 2 \log x, \log^2 x\}$ precisely for the paper’s even-trace family $x = n + \sqrt{n^2 \pm 1}$, and *not* for a generic quadratic unit such as the golden ratio—confirming, numerically, the paper’s remark that the individual values have no general closed form.

1 The function and its companion

The Herglotz(–Zagier) function may be taken in the integral form (the paper’s eq. (1))

$$J(x) = \int_0^1 \frac{\log(1+t^x)}{1+t} dt, \quad (1)$$

related to the Herglotz function $F(x)$ by (eq. (2))

$$J(x) = F(2x) - 2F(x) + F(x/2) + \frac{\pi^2}{12x}, \quad (2)$$

and the arithmetic identities of §6 are stated for the normalised companion

$$\mathcal{J}(x) = J(x) - \frac{\log^2 2}{2} + \frac{\pi^2}{24} \left(x - \frac{1}{x}\right), \quad \mathcal{J}(x) = \mathcal{F}(2x) - 2\mathcal{F}(x) + \mathcal{F}(x/2). \quad (3)$$

The arithmetic atoms are $S_{p,q} = \log \varepsilon_p \log \varepsilon_q$, where, for a fundamental discriminant $d > 1$, ε_d is the fundamental unit of $\mathbb{Q}(\sqrt{d})$, and $\varepsilon_1 := 2$. We implemented (??)–(??) as `intnum` integrals and $S_{p,q} = \mathbf{bnfinit}(x^2 - \mathbf{core}(d)).\mathbf{reg}$ products, and confirmed (??)–(??) together with the paper’s eqs. (4), (23), (24) to $\sim 10^{-20}$ before touching the tables.

2 Row-by-row verification

Tables 1 and 2 list, for the 45 known one-class-per-genus cases, the quantities

$$4\left(\mathcal{J}(n + \sqrt{n^2 - 1}) - n\frac{\pi^2}{24}\right) \quad (\text{Table 1}), \quad 2\left(\mathcal{J}(n + \sqrt{n^2 + 1}) - n\frac{\pi^2}{24}\right) \quad (\text{Table 2})$$

as integer combinations of the $S_{p,q}$. Evaluating both sides at 200 digits, all 16 rows of Table 2 (the identities the authors flag as “apparently new”) and 27 of the 29 rows of Table 1 agree to better than 10^{-80} . Two Table 1 rows fail, each by an $O(1)$ residual:

$$n = 43: \text{LHS} - \text{RHS} = -13.0160\dots, \quad n = 61: \text{LHS} - \text{RHS} = -2.9011\dots$$

3 The two errata

Both residuals resolve to a single, exact, structurally forced correction.

$n = 43$ ($n^2 - 1 = 2^3 \cdot 3 \cdot 7 \cdot 11 = 1848$). The printed right-hand side is

$$3S_{1,1848} + 2S_{8,924} + S_{12,616} + S_{24,77} + S_{28,264} + S_{44,168} + \mathbf{S_{33,56}} + S_{21,88}.$$

Its excess over the true value is *exactly* $S_{33,56}$: numerically $(\text{LHS} - \text{RHS})/S_{33,56} = -1.000\dots$ to 110 digits. A single-edit search (try deleting each term, changing each coefficient in $[-4, 4]$, or replacing each subscript pair by any fundamental-discriminant pair whose product is 1848 or $4 \cdot 1848$) returns *one* fix: delete the spurious $+S_{33,56}$. The remaining seven terms then satisfy the identity to 10^{-80} .

$n = 61$ ($n^2 - 1 = 2^3 \cdot 3 \cdot 5 \cdot 31 = 3720$). The printed right-hand side is

$$3S_{1,3720} + S_{12,1240} + 2S_{5,744} + S_{24,620} + 2S_{40,93} + \mathbf{S_{60,1032}} + S_{120,124}.$$

Here $1032 = 2^3 \cdot 3 \cdot 43$ carries the prime 43, which divides neither $n^2 - 1 = 3720$ nor $4(n^2 - 1)$; the subscript is impossible. The single-edit search returns the unique fix $S_{60,1032} \rightarrow S_{60,248}$, with $248 = 2^3 \cdot 31$ and $60 \cdot 248 = 14880 = 4 \cdot 3720$ a genuine genus splitting. The corrected row verifies to 10^{-80} .

With both corrections, **all 45 rows verify to better than 10^{-80}** . The high-DPI scan confirms that the printed paper itself carries both errors (they are not OCR artefacts of our transcription); our `HerglotzFunction.tex` reproduces the printed values faithfully and catalogues the corrections in an editorial note, per the repository’s transcription convention.

4 The elementary boundary

The closed forms in Tables 1–2 exist because $\mathcal{J} = \mathcal{F}(2x) - 2\mathcal{F}(x) + \mathcal{F}(x/2)$ is, at a unit $x = n + \sqrt{n^2 \pm 1}$, a combination of Kronecker limits whose non-elementary parts cancel. This is special to the *even-trace* units. Testing the three-term ansatz

$$\mathcal{J}(x) \stackrel{?}{=} a\zeta(2) + b\log 2 \log x + c\log^2 x$$

by integer-relation search (PARI `linddep`) we find it holds for the paper’s family, e.g.

$$\mathcal{J}(1 + \sqrt{2}) = \frac{1}{4}\zeta(2) + \frac{1}{2}\log 2 \log(1 + \sqrt{2}), \quad \mathcal{J}(2 + \sqrt{3}) = \frac{1}{2}\zeta(2) + \frac{1}{2}\log 2 \log(2 + \sqrt{3}),$$

but *fails* for every generic quadratic unit we tried—the golden ratio ϕ , ϕ^2 , the bronze unit $\frac{3+\sqrt{13}}{2}$, $\frac{5+\sqrt{21}}{2}$, $3 + \sqrt{10}$ —where `lindep` returns coefficients of size $\sim 10^{25}$ (no relation), despite $\mathbb{Q}(\sqrt{5})$ and the others having class number one. The golden value $\mathcal{J}(\phi)$ is therefore a genuinely deeper constant. This is the precise numerical content of the paper’s closing remark that “there are no similar formulas for $\mathcal{J}(\varepsilon)$ when ε is a unit of odd trace,” and that “there does not seem to be a general formula for the individual values $\mathcal{F}(x)$.”

Reproducibility

The integral/regulator implementation, the full 45-row verification (with the two corrections), the single-edit error localisation, and the elementary-boundary `lindep` tests are in `src/PolyLog/tools/scratch/herglotz/` (PARI/GP and `mpmath`); the verified statements and both errata are recorded in `src/PolyLog/identities/herglotz-quadratic-units.wl`.